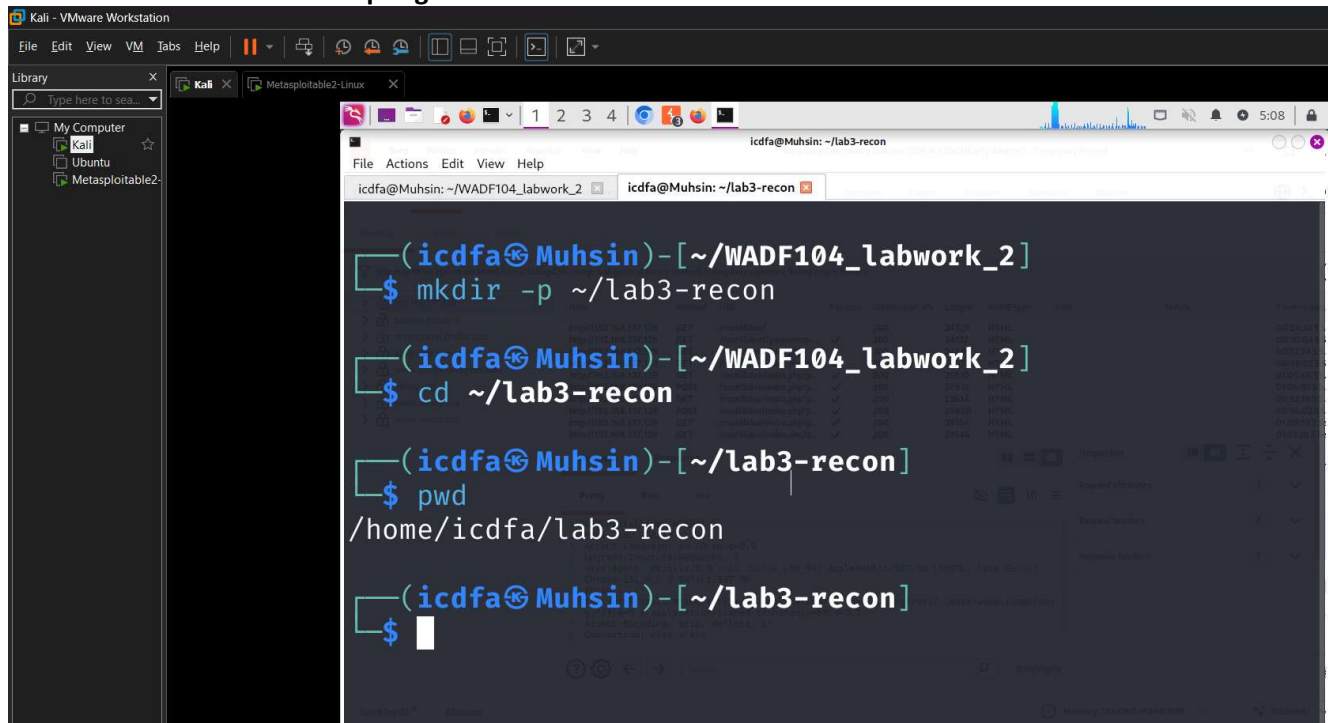


ICDFA
Linux I
WADF-2026-M04 Assignment

FULLNAME: SULEIMAN ABBEY BELLO
REGISTRATION NUMBER: C11/26/FCDF/17136
LINUX DISTRIBUTION/VERSION: ICDFA Kali Linux 2025.1
VM HOSTNAME: VMware Workstation Pro
WEEK: 10 (TEN) ASSIGNMENT
LAB 3 ASSESSMENT: Bash Scripting - Build a Reconnaissance Tool



The screenshot shows a Kali Linux terminal window with the following commands and output:

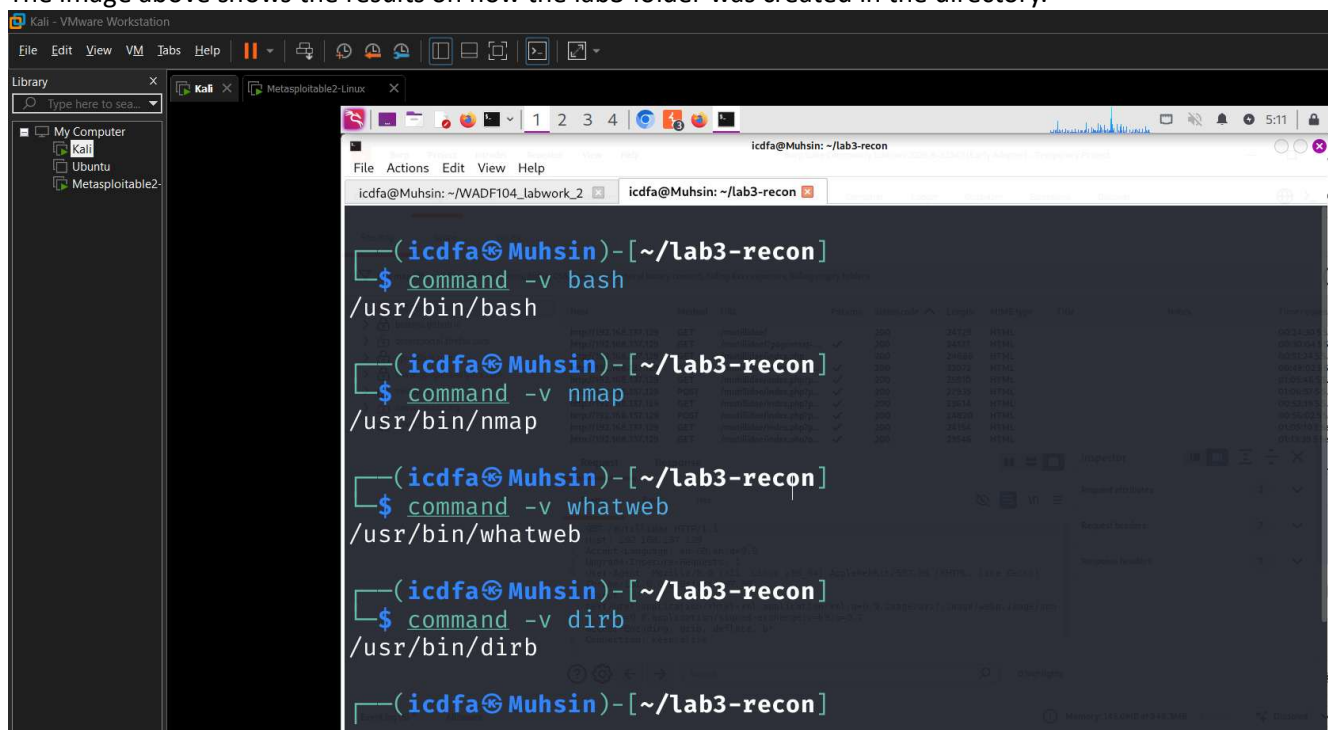
```
(icdfa@Muhsin)-[~/WADF104_labwork_2]
$ mkdir -p ~/lab3-recon

(icdfa@Muhsin)-[~/WADF104_labwork_2]
$ cd ~/lab3-recon

(icdfa@Muhsin)-[~/lab3-recon]
$ pwd
/home/icdfa/lab3-recon

(icdfa@Muhsin)-[~/lab3-recon]
$
```

The image above shows the results on how the lab3 folder was created in the directory.



The screenshot shows a Kali Linux terminal window with the following commands and output:

```
(icdfa@Muhsin)-[~/lab3-recon]
$ command -v bash
/usr/bin/bash

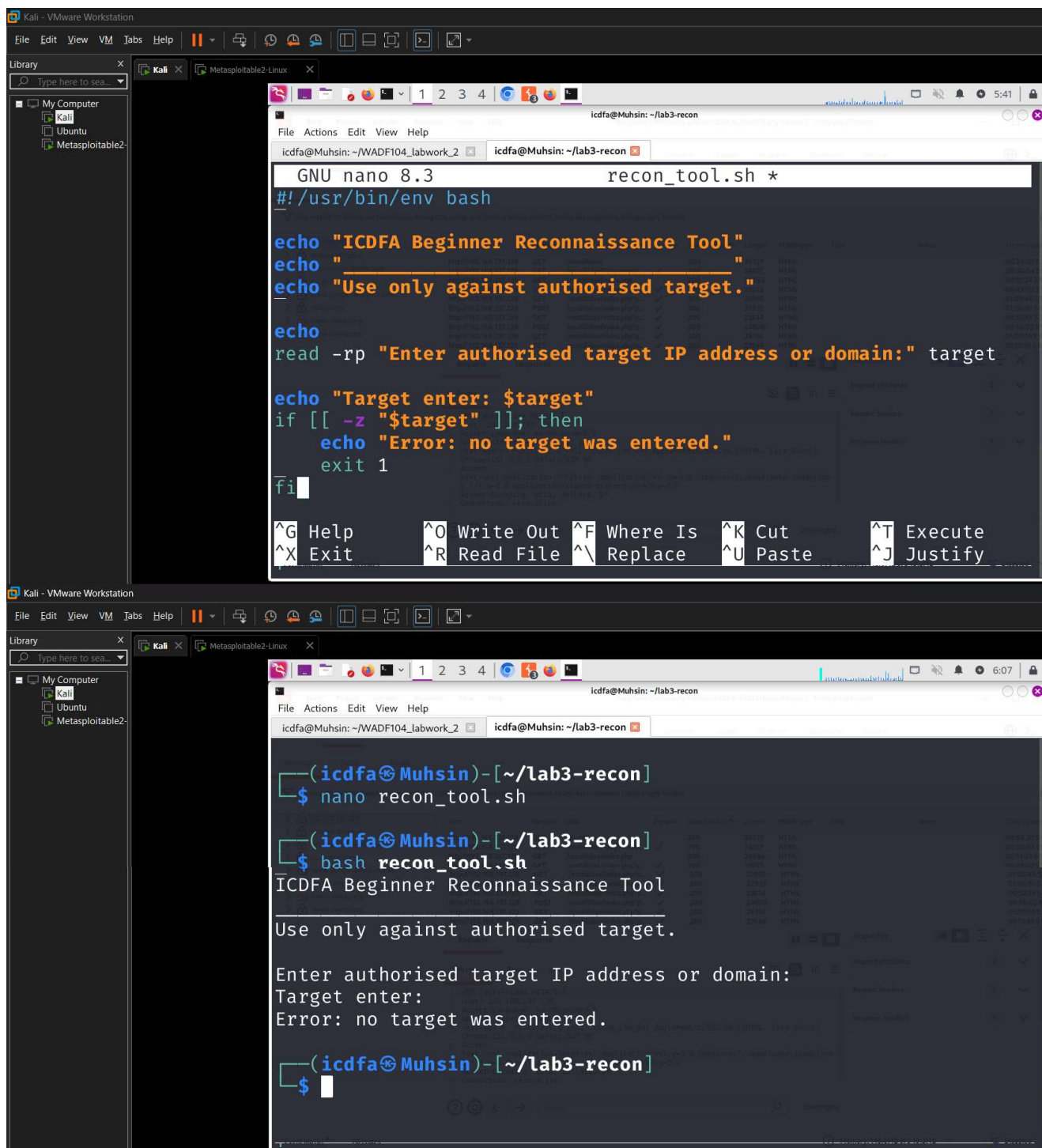
(icdfa@Muhsin)-[~/lab3-recon]
$ command -v nmap
/usr/bin/nmap

(icdfa@Muhsin)-[~/lab3-recon]
$ command -v whatweb
/usr/bin/whatweb

(icdfa@Muhsin)-[~/lab3-recon]
$ command -v dirb
/usr/bin/dirb

(icdfa@Muhsin)-[~/lab3-recon]
$
```

The image above shows the results on how the required tools were checks/confirmed.



The image above shows the result of building the scripts in stages from stage A-C.

```
Kali - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Kali
Ubuntu
Metasploitable2-
Kali x Metasploitable2-Linux x
icdfa@Muhsin: ~/lab3-recon
File Actions Edit View Help
icdfa@Muhsin: ~/WADFI04_labwork_2 x icdfa@Muhsin: ~/lab3-recon x
(icdfa@Muhsin)-[~/lab3-recon]
$ cat recon_tool.sh

#!/usr/bin/env bash

echo "ICDFA Beginner Reconnaissance Tool"
echo "Use only against authorised target."

echo
read -rp "Enter authorised target IP address or domain:" target

echo "Target enter: $target"
if [[ -z "$target" ]]; then
    echo "Error: no target was entered."
    exit 1
fi
echo
echo "Select a Reconnaissance tool:"
echo "1) Whatweb"
echo "2) Nmap"
```

```
Kali - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Kali
Ubuntu
Metasploitable2-
Kali x Metasploitable2-Linux x
icdfa@Muhsin: ~/lab3-recon
File Actions Edit View Help
icdfa@Muhsin: ~/WADFI04_labwork_2 x icdfa@Muhsin: ~/lab3-recon x
echo "3) DIRB"
echo "4) exit"
read -rp "Enter your choice [1-4]: " choice

check_tool() {
if ! command -v "$1" >/dev/null 2>&1; then
    echo "Error: required tool '$1' is not installed or not in PATH."
    exit 1
fi
}
case "$choice" in
1)
    check_tool whatweb
    echo "[+] Running Whatweb against $target"
    whatweb "http://$target"
    ;;
2)
    check_tool Nmap
    echo "[+] Running Nmap service detection against $target"
```



```
Kali - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Kali
Ubuntu
Metasploitable2
Kali x Metasploitable2-Linux x
icdfa@Muhsin: ~/lab3-recon
File Actions Edit View Help
icdfa@Muhsin: ~/WADFI04_labwork_2 x icdfa@Muhsin: ~/lab3-recon x
(icdfa@Muhsin)-[~/lab3-recon]
$ ./recon_tool.sh
ICDFA Beginner Reconnaissance Tool

Use only against authorised target.

Enter authorised target IP address or domain:192.168.137.129
Target enter: 192.168.137.129

Select a Reconnaissance tool:
1) Whatweb
2) Nmap
3) DIRB
4) exit
Enter your choice [1-4]: ^C

(icdfa@Muhsin)-[~/lab3-recon]
```

The image above shows the results for step 6.

```
Kali - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Kali
Ubuntu
Metasploitable2
Kali x Metasploitable2-Linux x
icdfa@Muhsin: ~/lab3-recon
File Actions Edit View Help
icdfa@Muhsin: ~/WADFI04_labwork_2 x icdfa@Muhsin: ~/lab3-recon x
Use only against authorised target.

Enter authorised target IP address or domain:192.168.137.129
Target enter: 192.168.137.129

Select a Reconnaissance tool:
1) Whatweb
2) Nmap
3) DIRB
4) exit
Enter your choice [1-4]: ^C

(icdfa@Muhsin)-[~/lab3-recon]
$ head -1 recon_tool.sh
#!/usr/bin/env bash

(icdfa@Muhsin)-[~/lab3-recon]
$
```



```
Kali - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Kali
Ubuntu
Metasploitable2
Kali x Metasploitable2-Linux x
icdfa@Muhsin: ~/lab3-recon
File Actions Edit View Help
icdfa@Muhsin: ~/WADFI04_labwork_2 x icdfa@Muhsin: ~/lab3-recon x
$ ./recon_tool.sh
ICDFA Beginner Reconnaissance Tool
Use only against authorised target.
Enter authorised target IP address or domain:192.168.137.129
Target enter: 192.168.137.129
Select a Reconnaissance tool:
1) Whatweb
2) Nmap
3) DIRB
4) exit
Enter your choice [1-4]: 4
Exiting. No scan was run.
(icdfa@Muhsin)-[~/lab3-recon]
$
```

```
Kali - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Kali
Ubuntu
Metasploitable2
Kali x Metasploitable2-Linux x
icdfa@Muhsin: ~/lab3-recon
File Actions Edit View Help
icdfa@Muhsin: ~/WADFI04_labwork_2 x icdfa@Muhsin: ~/lab3-recon x
(icdfa@Muhsin)-[~/lab3-recon]
$ ./recon_tool.sh
ICDFA Beginner Reconnaissance Tool
Use only against authorised target.
Enter authorised target IP address or domain:192.168.137.129
Target enter: 192.168.137.129
Select a Reconnaissance tool:
1) Whatweb
2) Nmap
3) DIRB
4) exit
Enter your choice [1-4]: 1
[+] Running Whatweb against 192.168.137.129
http://192.168.137.129 [200 OK] Apache[2.2.8], Country[RESERVED][Z],
HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], IP[192.168.137.129],
PHP[5.2.4-2ubuntu5.10], Title[Metasploitable2 - Linux], WebDAV[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]
```

```
Kali - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Kali
Ubuntu
Metasploitable2-
icdfa@Muhsin: ~/lab3-recon
File Actions Edit View Help
icdfa@Muhsin: ~/WADFI04_labwork_2 icdfa@Muhsin: ~/lab3-recon
(icdfa@Muhsin)-[~/lab3-recon]
$ ./recon_tool.sh
ICDFA Beginner Reconnaissance Tool

Use only against authorised target.

Enter authorised target IP address or domain:192.168.137.129
Target enter: 192.168.137.129

Select a Reconnaissance tool:
1) Whatweb
2) Nmap
3) DIRB
4) exit
Enter your choice [1-4]: 2
[+] Running Nmap service detection against 192.168.137.129
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-05 09:01 WAT
Nmap scan report for 192.168.137.129
Host is up (0.0066s latency).
Not shown: 977 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
```

```
Kali - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Kali
Ubuntu
Metasploitable2-
icdfa@Muhsin: ~/lab3-recon
File Actions Edit View Help
icdfa@Muhsin: ~/WADFI04_labwork_2 icdfa@Muhsin: ~/lab3-recon
23/tcp    open  telnet      Linux telnetd
25/tcp    open  smtp        Postfix smtpd
53/tcp    open  domain      ISC BIND 9.4.2
80/tcp    open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind     2 (RPC #100000)
139/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec?
513/tcp   open  login?
514/tcp   open  shell?
1099/tcp  open  java-rmi    GNU Classpath grmiregistry
1524/tcp  open  bindshell   Metasploitable root shell
2049/tcp  open  nfs         2-4 (RPC #100003)
2121/tcp  open  ccproxy-ftp?
3306/tcp  open  mysql?
5432/tcp  open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc         VNC (protocol 3.3)
6000/tcp  open  X11         (access denied)
6667/tcp  open  irc         UnrealIRCd
8009/tcp  open  ajp13       Apache Jserv (Protocol v1.3)
8180/tcp  open  unknown
```

```
Kali - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Kali
Ubuntu
Metasploitable2
Kali x Metasploitable2-Linux x
icdfa@Muhsin: ~/lab3-recon
File Actions Edit View Help
icdfa@Muhsin: ~/WADF104_labwork_2 x icdfa@Muhsin: ~/lab3-recon x
514/tcp open shell?
1099/tcp open java-rmi GNU Classpath grmiregistry
1524/tcp open bindshell Metasploitable root shell
2049/tcp open nfs 2-4 (RPC #100003)
2121/tcp open ccproxy-ftp?
3306/tcp open mysql?
5432/tcp open postgresql PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp open vnc VNC (protocol 3.3)
6000/tcp open X11 (access denied)
6667/tcp open irc UnrealIRCd
8009/tcp open ajp13 Apache Jserv (Protocol v1.3)
8180/tcp open unknown
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; O
Ss: Unix, Linux; CPE: cpe:/o:linux:linux_kernel
Service detection performed. Please report any incorrect results at https:/
/nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 178.17 seconds
(icdfa@Muhsin)-[~/lab3-recon]
$
```

```
Kali - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Kali
Ubuntu
Metasploitable2
Kali x Metasploitable2-Linux x
icdfa@Muhsin: ~/lab3-recon
File Actions Edit View Help
icdfa@Muhsin: ~/WADF104_labwork_2 x icdfa@Muhsin: ~/lab3-recon x
(icdfa@Muhsin)-[~/lab3-recon]
$ ./recon_tool.sh
ICDFA Beginner Reconnaissance Tool
Use only against authorised target.
Enter authorised target IP address or domain:192.168.137.129
Target enter: 192.168.137.129
Select a Reconnaissance tool:
1) Whatweb
2) Nmap
3) DIRB
4) exit
Enter your choice [1-4]: 3
./recon_tool.sh: line 41: Check_tool: command not found
[+] Running DIRB against 192.168.137.129
DIRB v2.22
By The Dark Raver
```



```
Kali - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Kali
Ubuntu
Metasploitable2
Kali x Metasploitable2-Linux x
icdfa@Muhsin: ~/lab3-recon
START_TIME: Sat Sep 5 09:07:15 2026
URL_BASE: http://192.168.137.129/
WORDLIST_FILES: /usr/share/dirb/wordlists/common.txt
GENERATED WORDS: 4612
Scanning URL: http://192.168.137.129/
+ http://192.168.137.129/cgi-bin/ (CODE:403|SIZE:296)
=> DIRECTORY: http://192.168.137.129/dav/
+ http://192.168.137.129/index (CODE:200|SIZE:891)
+ http://192.168.137.129/index.php (CODE:200|SIZE:891)
+ http://192.168.137.129/phpinfo (CODE:200|SIZE:48101)
+ http://192.168.137.129/phpinfo.php (CODE:200|SIZE:48113)
=> DIRECTORY: http://192.168.137.129/phpMyAdmin/
+ http://192.168.137.129/server-status (CODE:403|SIZE:301)
=> DIRECTORY: http://192.168.137.129/test/
```

```
Kali - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Kali
Ubuntu
Metasploitable2
Kali x Metasploitable2-Linux x
icdfa@Muhsin: ~/lab3-recon
Entering directory: http://192.168.137.129/dav/
(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)
Entering directory: http://192.168.137.129/phpMyAdmin/
+ http://192.168.137.129/phpMyAdmin/central (CODE:200|SIZE:4145)
+ http://192.168.137.129/phpMyAdmin/changelog (CODE:200|SIZE:74593)
+ http://192.168.137.129/phpMyAdmin/ChangeLog (CODE:200|SIZE:40540)
=> DIRECTORY: http://192.168.137.129/phpMyAdmin/contrib/
+ http://192.168.137.129/phpMyAdmin/docs (CODE:200|SIZE:4583)
+ http://192.168.137.129/phpMyAdmin/error (CODE:200|SIZE:1063)
+ http://192.168.137.129/phpMyAdmin/export (CODE:200|SIZE:4145)
+ http://192.168.137.129/phpMyAdmin/favicon.ico (CODE:200|SIZE:18902)
+ http://192.168.137.129/phpMyAdmin/import (CODE:200|SIZE:4145)
+ http://192.168.137.129/phpMyAdmin/index (CODE:200|SIZE:4145)
+ http://192.168.137.129/phpMyAdmin/index.php (CODE:200|SIZE:4145)
=> DIRECTORY: http://192.168.137.129/phpMyAdmin/js/
=> DIRECTORY: http://192.168.137.129/phpMyAdmin/lang/
```

```
Kali - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Kali
Ubuntu
Metasploitable2

icdfa@Muhsin: ~/lab3-recon
File Actions Edit View Help
icdfa@Muhsin: ~/WADF104_labwork_2 icdfa@Muhsin: ~/lab3-recon
+ http://192.168.137.129/phpMyAdmin/navigation (CODE:200|SIZE:4145)
+ http://192.168.137.129/phpMyAdmin/phpinfo (CODE:200|SIZE:0)
+ http://192.168.137.129/phpMyAdmin/phpinfo.php (CODE:200|SIZE:0)
+ http://192.168.137.129/phpMyAdmin/phpmyadmin (CODE:200|SIZE:21389)
+ http://192.168.137.129/phpMyAdmin/print (CODE:200|SIZE:1063)
+ http://192.168.137.129/phpMyAdmin/readme (CODE:200|SIZE:2624)
+ http://192.168.137.129/phpMyAdmin/README (CODE:200|SIZE:2624)
+ http://192.168.137.129/phpMyAdmin/robots (CODE:200|SIZE:26)
+ http://192.168.137.129/phpMyAdmin/robots.txt (CODE:200|SIZE:26)
=> DIRECTORY: http://192.168.137.129/phpMyAdmin/scripts/
=> DIRECTORY: http://192.168.137.129/phpMyAdmin/setup/
+ http://192.168.137.129/phpMyAdmin/sql (CODE:200|SIZE:4145)
=> DIRECTORY: http://192.168.137.129/phpMyAdmin/test/
=> DIRECTORY: http://192.168.137.129/phpMyAdmin/themes/
+ http://192.168.137.129/phpMyAdmin/TODO (CODE:200|SIZE:235)
+ http://192.168.137.129/phpMyAdmin/webapp (CODE:200|SIZE:6903)
— Entering directory: http://192.168.137.129/test/ —
(!) WARNING: Directory IS LISTABLE. No need to scan it.
```

```
Kali - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Kali
Ubuntu
Metasploitable2

icdfa@Muhsin: ~/lab3-recon
File Actions Edit View Help
icdfa@Muhsin: ~/WADF104_labwork_2 icdfa@Muhsin: ~/lab3-recon
— Entering directory: http://192.168.137.129/twiki/ —
=> DIRECTORY: http://192.168.137.129/twiki/bin/
+ http://192.168.137.129/twiki/data (CODE:403|SIZE:298)
+ http://192.168.137.129/twiki/index (CODE:200|SIZE:782)
+ http://192.168.137.129/twiki/index.html (CODE:200|SIZE:782)
=> DIRECTORY: http://192.168.137.129/twiki/lib/
+ http://192.168.137.129/twiki/license (CODE:200|SIZE:19440)
=> DIRECTORY: http://192.168.137.129/twiki/pub/
+ http://192.168.137.129/twiki/readme (CODE:200|SIZE:4334)
+ http://192.168.137.129/twiki/templates (CODE:403|SIZE:303)
— Entering directory: http://192.168.137.129/phpMyAdmin/contrib/ —
(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)
— Entering directory: http://192.168.137.129/phpMyAdmin/js/ —
(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)
```

```
icdfa@Muhsin: ~/lab3-recon
File Actions Edit View Help
icdfa@Muhsin: ~/WADFI04_labwork_2 icdfa@Muhsin: ~/lab3-recon

— Entering directory: http://192.168.137.129/phpMyAdmin/libraries/ —
(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

— Entering directory: http://192.168.137.129/phpMyAdmin/scripts/ —
(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

— Entering directory: http://192.168.137.129/phpMyAdmin/setup/ —
+ http://192.168.137.129/phpMyAdmin/setup/config (CODE:303|SIZE:1370)
⇒ DIRECTORY: http://192.168.137.129/phpMyAdmin/setup/frames/
+ http://192.168.137.129/phpMyAdmin/setup/index (CODE:200|SIZE:8619)
+ http://192.168.137.129/phpMyAdmin/setup/index.php (CODE:200|SIZE:8627)
⇒ DIRECTORY: http://192.168.137.129/phpMyAdmin/setup/lib/
+ http://192.168.137.129/phpMyAdmin/setup/scripts (CODE:200|SIZE:21967)
+ http://192.168.137.129/phpMyAdmin/setup/styles (CODE:200|SIZE:6218)

— Entering directory: http://192.168.137.129/phpMyAdmin/test/ —

icdfa@Muhsin: ~/lab3-recon
File Actions Edit View Help
icdfa@Muhsin: ~/WADFI04_labwork_2 icdfa@Muhsin: ~/lab3-recon

— Entering directory: http://192.168.137.129/twiki/lib/ —
(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

— Entering directory: http://192.168.137.129/twiki/pub/ —
(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

— Entering directory: http://192.168.137.129/phpMyAdmin/setup/frames/ --
--
(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

— Entering directory: http://192.168.137.129/phpMyAdmin/setup/lib/ —
(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

END_TIME: Sat Sep 5 09:07:56 2026
DOWNLOADED: 18448 - FOUND: 42
```

The image above shows the results for mandatory tools test. The reconnaissance phase was conducted against the authorised Metasploitable 2 laboratory target using Nmap, WhatWeb and DIRB. Nmap was used to identify open ports, running services and available version information. The scan revealed several exposed services, including HTTP, SSH, Telnet, SMTP, DNS, SMB, MySQL, PostgreSQL, VNC and other network services. WhatWeb was then used to fingerprint the technologies exposed by the web server and provided information about the web technologies and software supporting the application. DIRB was used for web content and directory discovery and successfully identified **42 resources**, including directories and files such as **/dav/**, **/phpMyAdmin/**, **/test/**, **/twiki/** and **index.php**. Different HTTP response codes, including 200, 301 and 403, were observed during the directory discovery process.


```
Kali - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Kali
Ubuntu
Metasploitable2-
icdfa@Muhsin: ~/lab3-recon
icdfa@Muhsin: ~/lab3-recon
icdfa@Muhsin: ~/lab3-recon
$ cd ~/lab3-recon
icdfa@Muhsin: ~/lab3-recon
$ ls -l
total 4
-rwxrwxr-x 1 icdfa icdfa 1077 Sep  5 08:43 recon_tool.sh
icdfa@Muhsin: ~/lab3-recon
$ cat recon_tool.sh
#!/usr/bin/env bash
echo "ICDFA Beginner Reconnaissance Tool"
echo "
echo "Use only against authorised target."
echo
```

```
Kali - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Kali
Ubuntu
Metasploitable2-
icdfa@Muhsin: ~/lab3-recon
icdfa@Muhsin: ~/lab3-recon
read -rp "Enter authorised target IP address or domain:" target
echo "Target enter: $target"
if [[ -z "$target" ]]; then
    echo "Error: no target was entered."
    exit 1
fi
echo "Select a Reconnaissance tool:"
echo "1) Whatweb"
echo "2) Nmap"
echo "3) DIRB"
echo "4) exit"
read -rp "Enter your choice [1-4]: " choice
check_tool() {
if ! command -v "$1" >/dev/null 2>&1; then
    echo "Error: required tool '$1' is not installed or not in PATH."
}
```



```
Kali - VMware Workstation
File Edit View VM Tabs Help
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Type here to search
My Computer
Kali
Ubuntu
Metasploitable2
Kali x Metasploitable2-Linux x
icdfa@Muhsin: ~/lab3-recon
File Actions Edit View Help
icdfa@Muhsin: ~/WADFI04_labwork_2 x icdfa@Muhsin: ~/lab3-recon x
exit 1
fi
}
case "$choice" in
1)
check_tool whatweb
echo "[+] Running Whatweb against $target"
whatweb "http://$target"
;;
2)
check_tool nmap
echo "[+] Running Nmap service detection against $target"
nmap -sV "$target"
;;
3)
check_tool dirb
echo "[+] Running DIRB against $target"
dirb "http://$target"
;;
*)
echo "Error: invalid menu choice."
exit 1
;;
esac
```

```
Kali - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Kali
Ubuntu
Metasploitable2
Kali x Metasploitable2-Linux x
icdfa@Muhsin: ~/lab3-recon
File Actions Edit View Help
icdfa@Muhsin: ~/WADFI04_labwork_2 x icdfa@Muhsin: ~/lab3-recon x
Check_tool dirb
echo "[+] Running DIRB against $target"
dirb "http://$target"
;;
4)
echo "Exiting. No scan was run."
exit 0
;;
*)
echo "Error: invalid menu choice."
exit 1
;;
esac
(icdfa@Muhsin)-[~/lab3-recon]
$
```

```
Kali - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Kali
Ubuntu
Metasploit2
icdfa@Muhsin: ~/lab3-recon
File Actions Edit View Help
icdfa@Muhsin: ~/WADFI04_labwork_2 icdfa@Muhsin: ~/lab3-recon
(icdfa@Muhsin)-[~/lab3-recon]
$ head -n 25 recon_tool.sh
#!/usr/bin/env bash

echo "ICDFA Beginner Reconnaissance Tool"
echo "_____ "
echo "Use only against authorised target."

echo
read -rp "Enter authorised target IP address or domain:" target
t

echo "Target enter: $target"
if [[ -z "$target" ]]; then
    echo "Error: no target was entered."
    exit 1
```

```
Kali - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Kali
Ubuntu
Metasploit2
icdfa@Muhsin: ~/lab3-recon
File Actions Edit View Help
icdfa@Muhsin: ~/WADFI04_labwork_2 icdfa@Muhsin: ~/lab3-recon
exit 1
fi
echo
echo "Select a Reconnaissance tool:"
echo "1) Whatweb"
echo "2) Nmap"
echo "3) DIRB"
echo "4) exit"
read -rp "Enter your choice [1-4]: " choice

check_tool() {
if ! command -v "$1" >/dev/null 2>&1; then
    echo "Error: required tool '$1' is not installed or not in P
ATH."
}

(icdfa@Muhsin)-[~/lab3-recon]
```

```
(icdfa@Muhsin)-[~/lab3-recon]
$ head -n 10 recon_tool.sh
#!/usr/bin/env bash

echo "ICDFA Beginner Reconnaissance Tool"
echo "Use only against authorised target."

echo
read -rp "Enter authorised target IP address or domain:" target
t

echo "Target enter: $target"

(icdfa@Muhsin)-[~/lab3-recon]
$

(icdfa@Muhsin)-[~/lab3-recon]
$ ./recon_tool.sh
ICDFA Beginner Reconnaissance Tool
Use only against authorised target.

Enter authorised target IP address or domain:192.168.137.129
Target enter: 192.168.137.129

Select a Reconnaissance tool:
1) Whatweb
2) Nmap
3) DIRB
4) exit
Enter your choice [1-4]:
```

The images above shows for the Part 8. The Bash reconnaissance tool successfully accepts an authorised target IP address or domain from the user and displays a menu containing WhatWeb, Nmap, DIRB, and an exit option. The tool uses conditional logic to execute the selected reconnaissance tool. Successful execution of WhatWeb, Nmap, and DIRB was demonstrated against the authorised laboratory target.

Part 9: The reconnaissance tool was developed in Bash to automate basic reconnaissance against authorised laboratory targets. It accepts the target from the user, validates the input, presents a menu of reconnaissance tools, and uses a `case` statement to execute the selected tool. Nmap provides service and version information, WhatWeb identifies web technologies, and DIRB performs web-content discovery. The target is supplied at runtime rather than being hardcoded, making the tool reusable for authorised lab targets.

Part 10:

1. What does the shebang do?

Answer: The shebang specifies Bash as the interpreter that should execute the script.

2. What does read -rp do?

Answer: read -rp displays a prompt and accepts the user's input, storing the entered target in the `target` variable.

3. Why quote "\$target"?

Answer: "\$target" is quoted so that Bash treats the target value as a single argument and handles the input safely.

4. What does -z test?

Answer: The `-z` condition checks whether the target variable is empty. If no target is entered, the script displays an error and exits.

5. What is the purpose of exit 1?

Answer: exit 1 terminates the script and indicates that an error or unsuccessful condition occurred.

6. Why is case suitable for menus?

Answer: The case statement is used to handle the different menu options and execute the appropriate action based on the user's choice.

7. What does ;; mean?

Answer: ;; marks the end of a case option and prevents Bash from continuing into the next option.

8. What does * mean in the case statement?

Answer: The * is the default case. It handles any input that does not match the available menu options.

9. Why does WhatWeb/DIRB use http://\$target?

Answer: WhatWeb and DIRB use http://\$target because they perform reconnaissance against web applications and need an HTTP URL to connect to the target's web server.

10. What does nmap -sV do?

Answer: nmap -sV performs service and version detection, helping identify the services running on open ports and their software versions.

11. What does command -v check?

Answer: command -v checks whether a required command or tool is available in the system's PATH before the script attempts to use it.

12. What does chmod +x do?

Answer: chmod +x gives the Bash script execute permission, allowing it to be run directly from the terminal.

13. What does ./ mean?

Answer: ./ specifies that the script should be executed from the current directory.

14. What does tee do?

Answer: tee allows command output to be displayed on the terminal while simultaneously saving the output to a file for later reference.

15. Which requirement prevents hardcoding?

Answer: The target is not hardcoded because the script uses the **read** command to accept the target IP address or domain from the user at runtime. This makes the tool reusable for different authorised laboratory targets.

Conclusion:

The Bash reconnaissance tool was successfully developed and tested against an authorised laboratory target. The script accepts a target at runtime and provides a menu for selecting WhatWeb, Nmap, or DIRB. The three tools were successfully executed, demonstrating basic automation of reconnaissance tasks using Bash scripting. The exercise also provided practical experience with Bash variables, conditional statements, functions, case statements, command execution, and file permissions.